

CSE370 : Database Systems
Project Report
Project Title : CareerBridge BRAC

Group No : 07, CSE370 Lab Section : 16, Fall 2025		
ID	**Name**	**Contribution**
24101184	TADRIB ISHAT	**Features worked on for FRONTEND :** 1.User Registration and Login System 2.Opportunity Posting and Management Functionality 3.Application Management System **Features worked on for BACKEND :** 1.User Registration and Login System 2.Opportunity Posting and Management Functionality 3.Application Management System
24101186	MD. ATIKULLAH TANVIR	**Features worked on for FRONTEND :** 1.Eligibility and Matching Analysis 2.Review and Probability Analysis System 3.Admin Dashboard and Control Panel **Features worked on for BACKEND :** 1.Eligibility and Matching Analysis 2.Review and Probability Analysis System 3.Admin Dashboard and Control Panel

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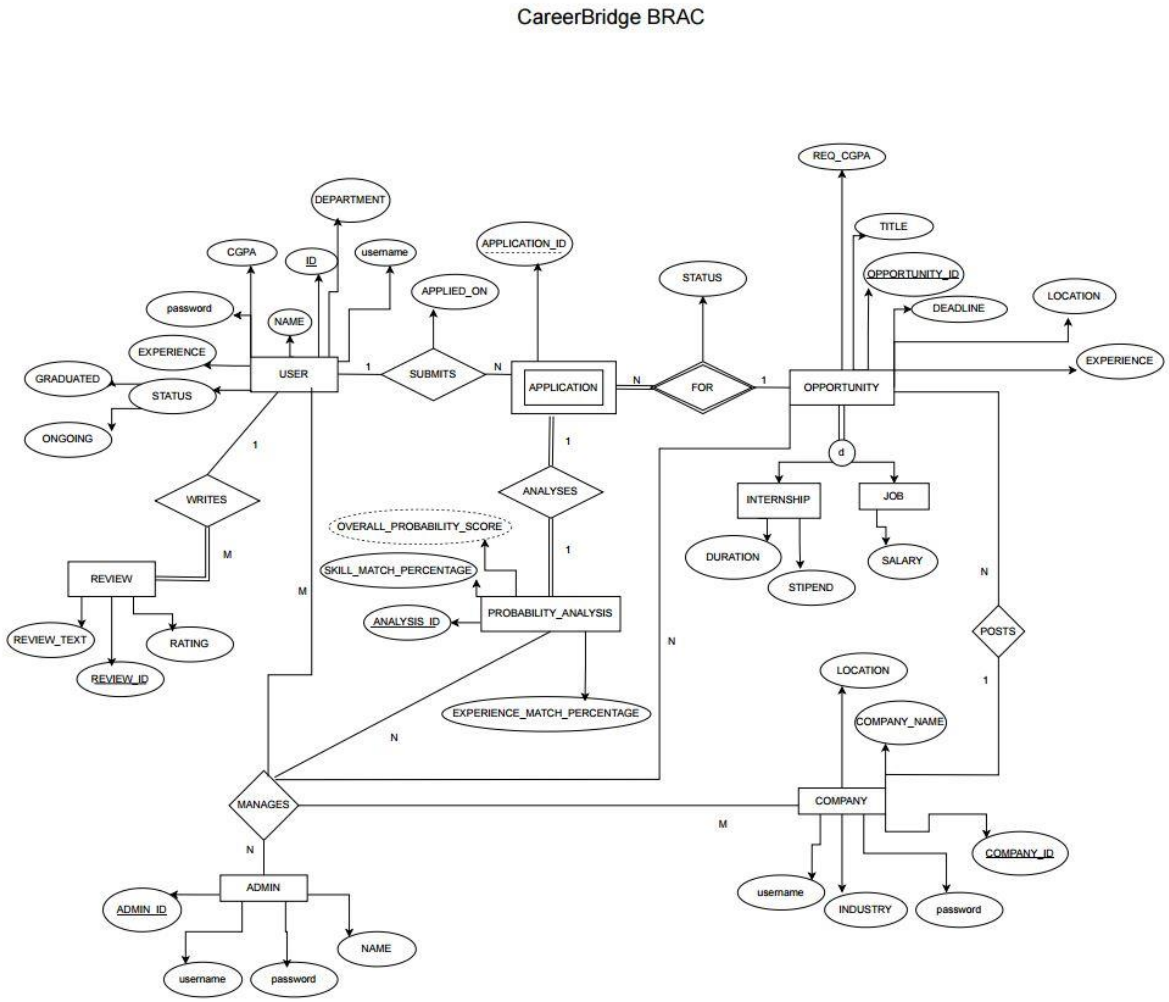
Introduction

CareerBridge BRAC is a PHP-based web application developed to facilitate an efficient and structured recruitment and career management process. The system is designed to connect students or job seekers with companies offering job and internship opportunities through a centralized digital platform. By automating major recruitment tasks, the project reduces manual processing and enhances transparency, accessibility, and decision-making accuracy.The system incorporates a **user registration and login system** to ensure secure access for users, companies, and administrators through session-based authentication. It provides **opportunity posting and management functionality**, allowing companies to publish and maintain detailed job or internship listings. Through the **application management system**, users can apply for opportunities and track their application status in real time.To improve candidate evaluation, the system implements **eligibility and matching analysis**, where user qualifications such as CGPA, skills, and experience are compared against opportunity requirements. A dedicated **review and probability analysis system** further processes these parameters to generate an overall probability score, supporting data-driven candidate shortlisting. Additionally, a comprehensive **admin dashboard and control panel** is included to manage users, companies, applications, and analytical operations efficiently.By integrating these features into a single web-based platform, CareerBridge BRAC demonstrates the practical application of PHP, database management, and web technologies in solving real-world recruitment challenges.

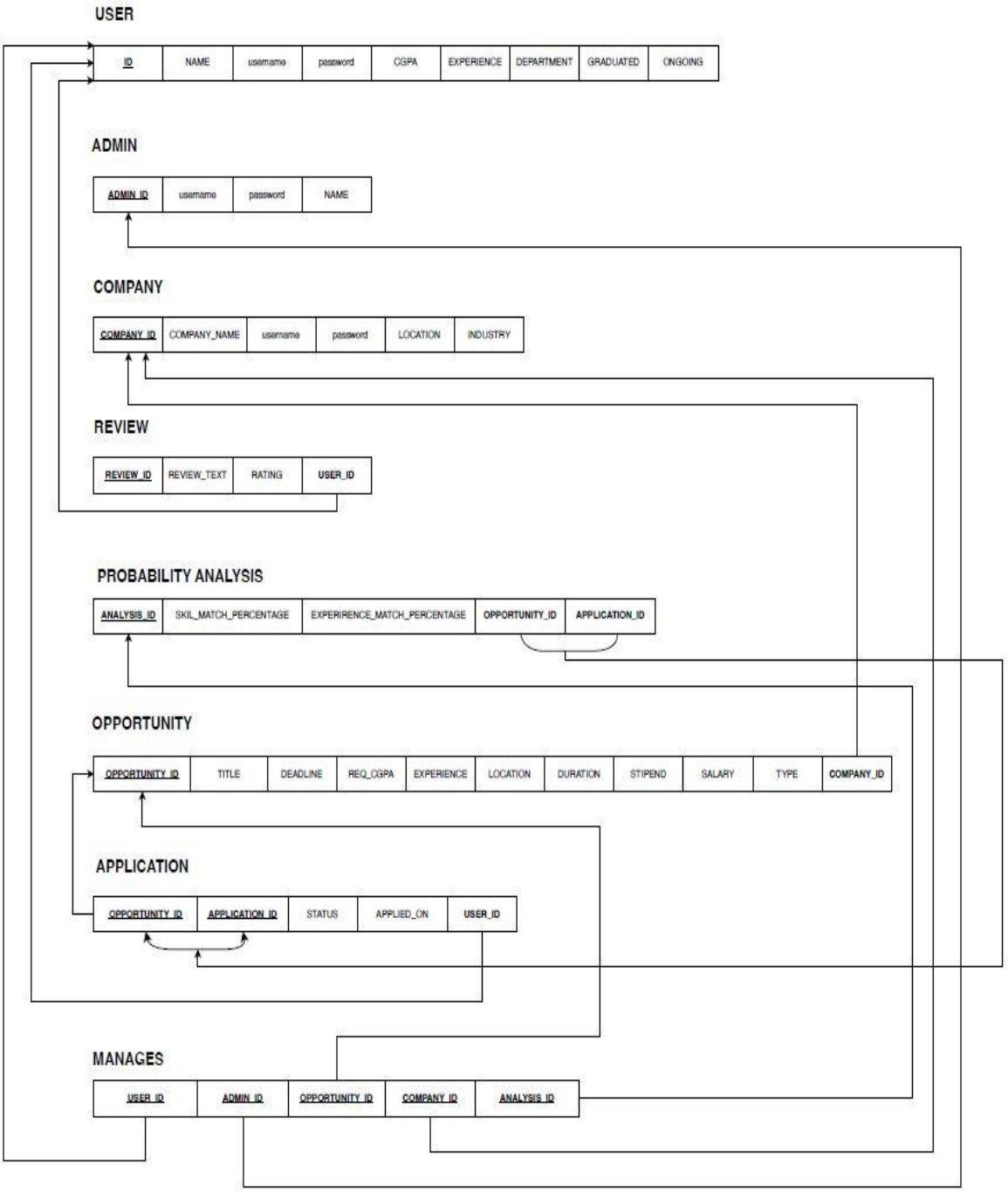
Project Features

ID, Name	Features	
24101184, TADRIB ISHAT	Ft 1	**User Registration and Login System** Designed frontend login/registration forms and implemented PHP-based authentication, session handling, and secure login/logout functionality.
	Ft 2	**Opportunity Posting and Management Functionality** Developed frontend forms and tables for opportunity management and implemented backend PHP logic for creating, updating, and deleting opportunities.
	Ft 3	**Application Management System** Designed application submission and tracking interfaces and implemented backend logic to store applications and manage their statuses.
24101186, MD. ATIKULLAH TANVIR	Ft 1	**Eligibility and Matching Analysis** Designed frontend displays for eligibility results and implemented PHP logic to evaluate CGPA, skills, and experience matching.
	Ft 2	**Review and Probability Analysis System** Developed frontend pages to show probability scores and implemented backend algorithms to calculate overall probability scores.
	Ft 3	**Admin Dashboard and Control Panel** Designed the admin dashboard interface and implemented backend PHP controls for managing users, companies, opportunities, and applications.

ER/EER Diagram



Schema Diagram



Normalization

a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.
b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?
c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?

ANSWER :

a. The converted schema satisfies 1NF as all relations contain a primary key and all attributes hold atomic, indivisible values. There are no repeating groups or multi-valued attributes. Therefore, no decomposition is required

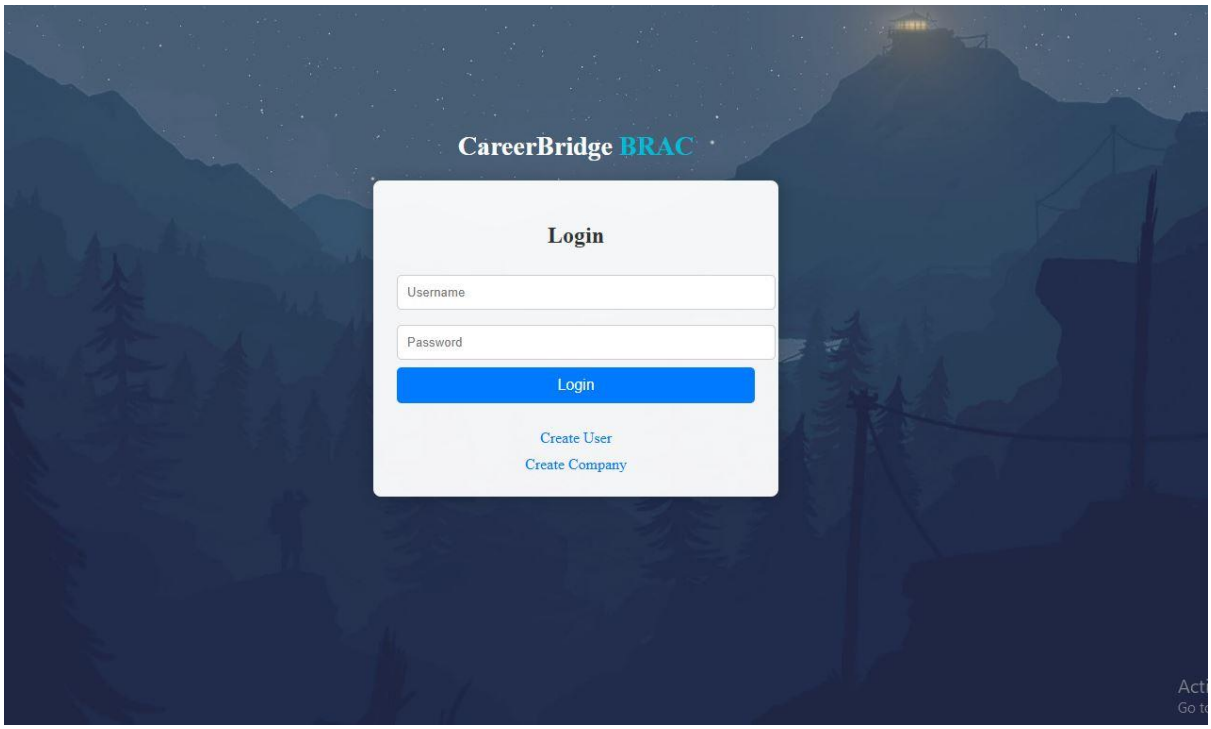
b. The schema is in 2NF because each relation uses a single-attribute primary key. As a result, partial functional dependencies cannot exist. Hence, no decomposition is necessary

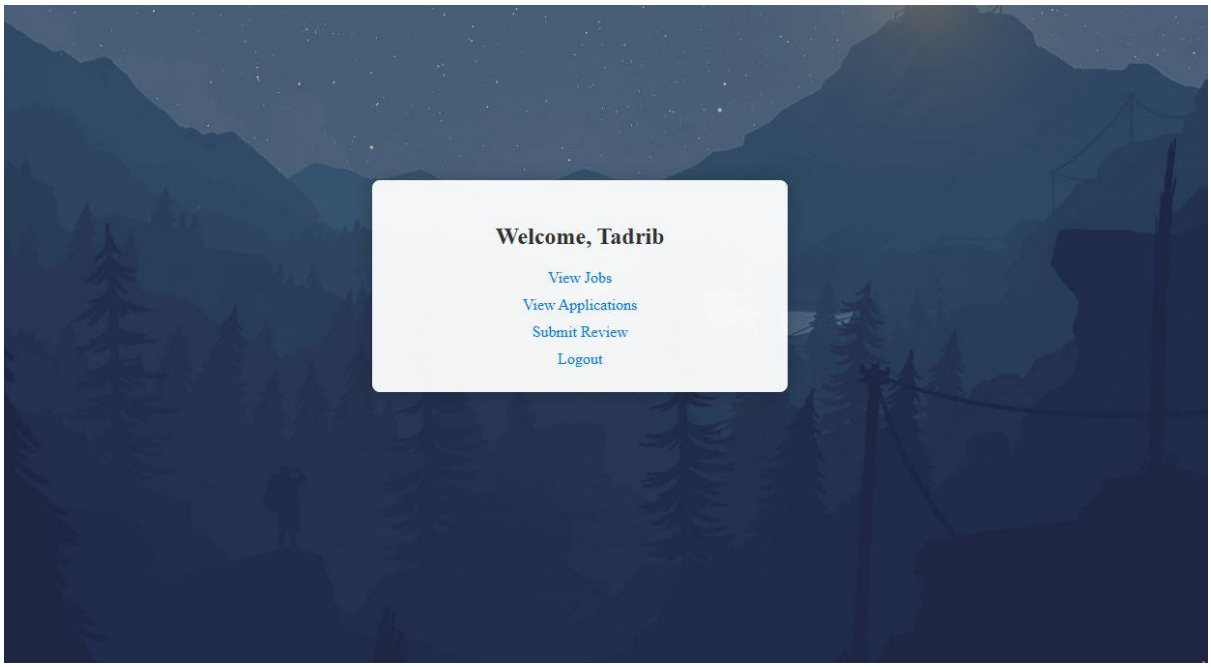
c. The schema also satisfies 3NF since all non-key attributes depend only on the primary key and not on other non-key attributes. There are no transitive dependencies present. Therefore, no further decomposition is required.

Frontend Development

Contribution of ID : 24101184 Name : TADRIB ISHAT

Designed and implemented the login and user registration interfaces, application-related pages, and ensured proper form layout and validation using HTML and CSS.

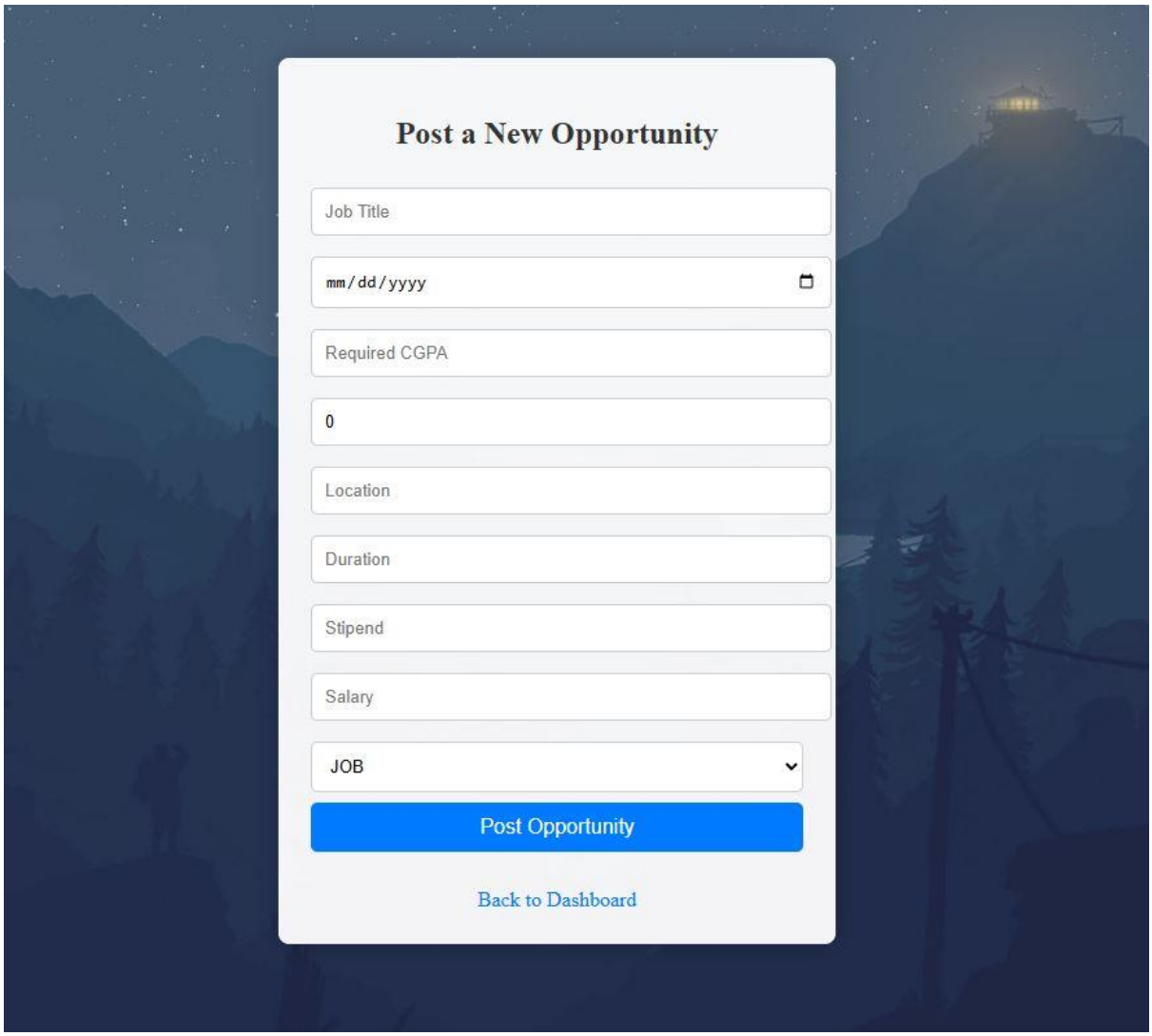


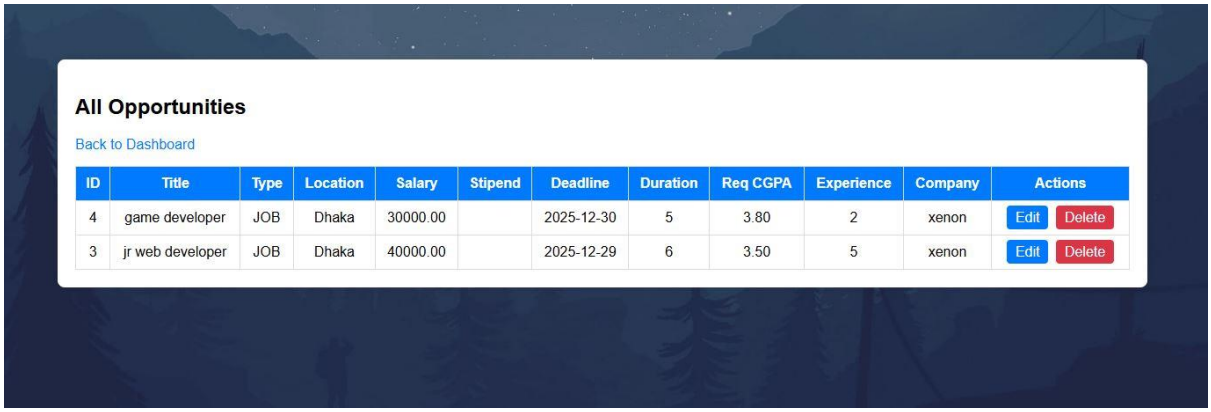


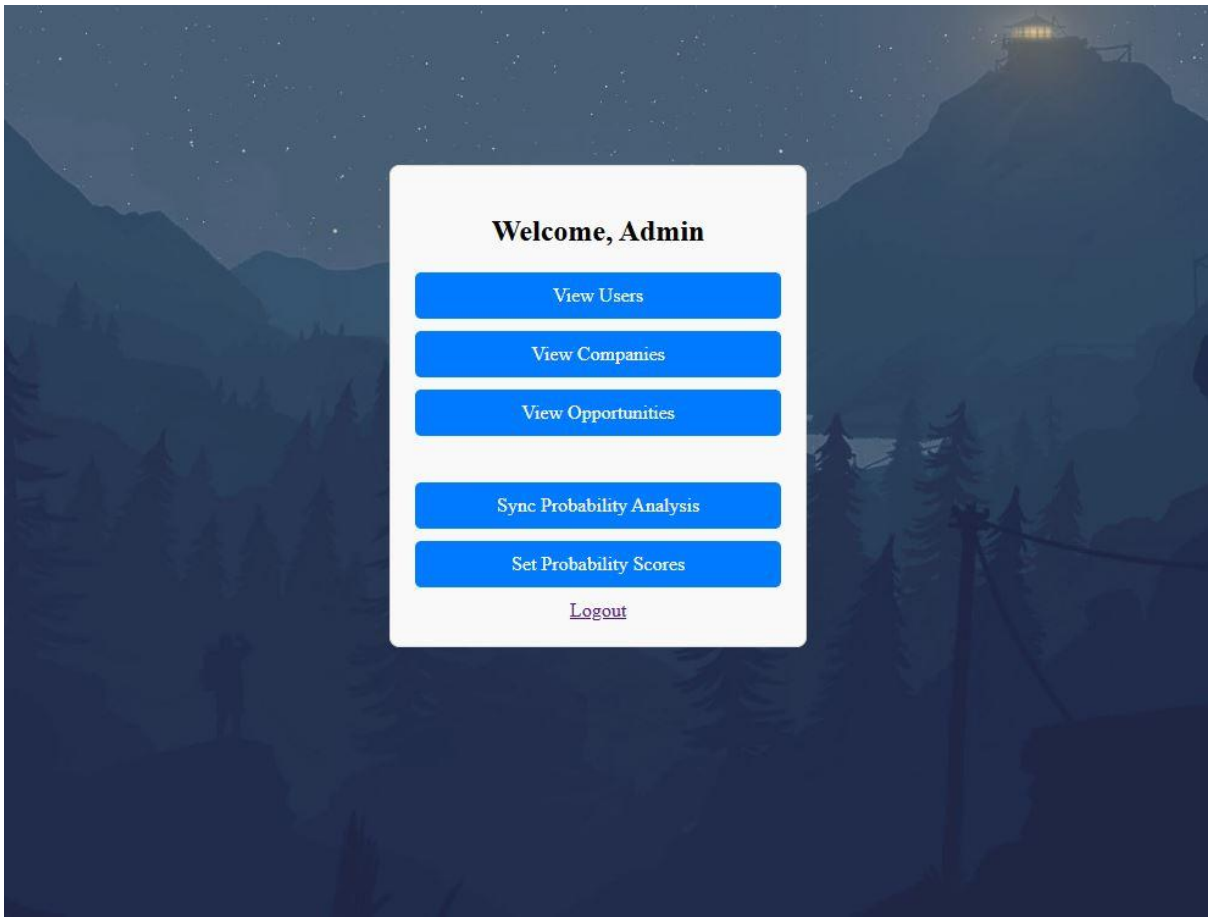
Frontend Development

Contribution of ID : 24101186 Name : MD. ATIKULLAH TANVIR

Developed the opportunity posting pages, admin dashboard layout, and styled tables and management panels to ensure consistency and usability.




All Opportunities
Back to Dashboard
ID Title Type Location Salary Stipend Deadline Duration Req CGPA Experience Company Actions
4 game developer JOB Dhaka 30000.00 2025-12-30 5 3.80 2 xenon Edit Delete
3 jr web developer JOB Dhaka 40000.00 2025-12-29 6 3.50 5 xenon Edit Delete


Welcome, Admin
View Users
View Companies
View Opportunities
Sync Probability Analysis
Set Probability Scores
Logout

Backend Development

Contribution of ID : 24101184 Name : TADRIB ISHAT

Implemented user registration & login system, application management, and eligibility & matching analysis using PHP.

```php
// ---------- CHECK COMPANY ----------
$stmt = $conn->prepare("SELECT company_id, company_name, password FROM company WHERE username=?");
$stmt->bind_param("s", $username);
$stmt->execute();
$res = $stmt->get_result();

if ($company = $res->fetch_assoc()) {
    if (password_verify($password, $company['password'])) {

        $_SESSION['company_id']   = $company['company_id'];
        $_SESSION['company_name'] = $company['company_name'];

        header("Location: company/dashboard.php");
        exit();
    }
}


// ---------- CHECK USER ----------
$stmt = $conn->prepare("SELECT ID as user_id, name, password FROM user WHERE username=?");
$stmt->bind_param("s", $username);
$stmt->execute();
$res = $stmt->get_result();

if ($user = $res->fetch_assoc()) {
    if (password_verify($password, $user['password'])) {

        $_SESSION['user_id'] = $user['user_id'];
        $_SESSION['name']    = $user['name'];

        header("Location: user/dashboard.php");
        exit();
    }
}

// ---------- CHECK ADMIN ----------
$admin_username = 'admin123';
$admin_password = 'admin@1122';
$admin_username = 'admin184';
$admin_password = 'admin@184';

if ($username === $admin_username && $password === $admin_password) {
    $_SESSION['admin_id'] = 1;
    $_SESSION['admin_name'] = 'Admin';
    header("Location: admin/dashboard.php");
    exit();
}

// ---------- FAILED ----------
header("Location: login.php?error=1");
exit();
```

```php
<?php
session_start();
session_destroy();
header("Location:login.php");
```

```
$sql = "
SELECT
    a.APPLICATION_ID,
    u.name AS user_name,
    o.TITLE AS job_title,
    p.SKILL_MATCH_PERCENTAGE,
    p.EXPERIENCE_MATCH_PERCENTAGE,
    p.OVERALL_PROBABILITY_SCORE
FROM application a
JOIN user u ON a.USER_ID = u.ID
JOIN opportunity o ON a.OPPORTUNITY_ID = o.OPPORTUNITY_ID
LEFT JOIN probability_analysis p ON a.APPLICATION_ID = p.APPLICATION_ID
ORDER BY a.APPLICATION_ID DESC
";
```

Backend Development

Contribution of ID : 24101186 Name : MD. ATIKULLAH TANVIR

Developed opportunity posting & management, review & probability analysis, and admin dashboard using PHP.

```php
<?php
session_start();
require '../DBconnect.php';

if (!isset($_SESSION['company_id'])) {
    header("Location: ../login.php");
    exit();
}

$success = $error = "";
$company_id = $_SESSION['company_id'];

if ($_SERVER['REQUEST_METHOD'] === 'POST') {

    $title = $_POST['title'];
    $deadline = $_POST['deadline'] ?: NULL;
    $req_cgpa = $_POST['req_cgpa'] ?: NULL;
    $experience = $_POST['experience'] ?: 0;
    $location = $_POST['location'] ?: NULL;
    $duration = $_POST['duration'] ?: NULL;
    $stipend = $_POST['stipend'] ?: NULL;
    $salary = $_POST['salary'] ?: NULL;
    $type = $_POST['type'];

    $sql = "INSERT INTO opportunity
        (TITLE, DEADLINE, REQ_CGPA, EXPERIENCE, LOCATION, DURATION, STIPEND, SALARY, TYPE, COMPANY_ID)
        VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?)";

    $stmt = $conn->prepare($sql);
    $stmt->bind_param(
        "ssdisdddssi",
        $title,
        $deadline,
        $req_cgpa,
        $experience,
        $location,
        $duration,
        $stipend,
        $salary,
        $type,
        $company_id
    );

    if ($stmt->execute()) {
        $success = "Job opportunity posted successfully!";
    } else {
        $error = "Error posting job: " . $stmt->error;
    }
}
?>
```

```php
// ------------------ FETCH APPLICATIONS ------------------
$sql = "
SELECT
    a.APPLICATION_ID,
    u.name AS user_name,
    o.TITLE AS job_title,
    p.SKILL_MATCH_PERCENTAGE,
    p.EXPERIENCE_MATCH_PERCENTAGE,
    p.OVERALL_PROBABILITY_SCORE
FROM application a
JOIN user u ON a.USER_ID = u.ID
JOIN opportunity o ON a.OPPORTUNITY_ID = o.OPPORTUNITY_ID
LEFT JOIN probability_analysis p ON a.APPLICATION_ID = p.APPLICATION_ID
ORDER BY a.APPLICATION_ID DESC
";

$result = $conn->query($sql);
?>
```

```php
<?php
session_start();
require '../DBconnect.php';

if (!isset($_SESSION['user_id'])) {
    header("Location: ../login.php");
    exit();
}

$user_id = $_SESSION['user_id'];
$message = "";

// Handle form submission
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $review_text = $_POST['review_text'];
    $rating = $_POST['rating'];

    $stmt = $conn->prepare("INSERT INTO review (ID, REVIEW_TEXT, RATING) VALUES (?, ?, ?)");
    $stmt->bind_param("isi", $user_id, $review_text, $rating);

    if ($stmt->execute()) {
        $message = "Thank you for your review!";
    } else {
        $message = "Error submitting review: " . $stmt->error;
    }
}
?>
```

Source Code Repository

DRIVE LINK : **CareerBridge BRAC**

Conclusion

CareerBridge BRAC successfully demonstrates how a PHP-based web application can effectively manage recruitment and career opportunities in a structured and efficient manner. The system integrates core functionalities such as user registration and login, opportunity posting and management, application management, and an admin dashboard with a centralized control panel. In addition, advanced features like eligibility and matching analysis and the review and probability analysis system were implemented at a later stage, further enhancing decision-making and candidate–opportunity alignment. Overall, the project successfully implements all intended features, showcasing the practical application of PHP, database management, and web development concepts while providing a scalable foundation for future enhancements such as advanced analytics, automated notifications.

References

PROJECT SETUP :
https://drive.google.com/drive/folders/1QXTZ120Wt1gJiTPMlM9cWfsgZZkiV8FQ?usp=sharing

PROJECT VIDEO :
https://www.youtube.com/playlist?list=PLJW6cU20q-SOQB1Dqj2vF9mcNqgh4-z_j
https://www.youtube.com/watch?v=d5Hf6d6QtIo&t=2962s
https://www.youtube.com/watch?v=J8QPm8_j6lM
https://www.youtube.com/watch?v=zZ6vybT1HQs